

Layer	Component	Open source tools
Model foundation	Base models	Mistral 7B, Mixtral 8x7B, Llama 3.3 70B, Phi-4, Gemma 2 (2B-27B), Qwen2.5 (0.5B-72B)
	Model hub	Hugging Face Hub, Ollama
Data pipeline	Data versioning	DVC, Git LFS, LakeFS
	Annotation	Label Studio, Argilla, Prodigy
	Synthetic data	NeMo Curator, DataTrove, Lilac
Training and fine-tuning	Training framework	Hugging Face Transformers, Axolotl, LLaMA-Factory, PEFT
	Distributed training	DeepSpeed, Accelerate, PyTorch FSDP
	Experiment tracking	Weights & Biases, MLflow, TensorBoard
Inference infrastructure	GPU inference	vLLM, TensorRT-LLM, Text Generation Inference (TGI), LocalAI
	CPU inference	llama.cpp, GGML/GGUF, ONNX Runtime
	Model serving	Ray Serve, BentoML, KServe, Triton Inference Server
	Quantization	bitsandbytes, GPTQ, AWQ, AutoGPTQ
Orchestration	Application framework	LangChain, LlamaIndex, Haystack, Semantic Kernel
	Vector database	Qdrant, Weaviate, Milvus, Chroma, pgvector
	Agent framework	AutoGen, CrewAI, LangGraph, OpenAI Swarm
Deployment	Containerisation	Docker, Podman
	Orchestration	Kubernetes, Docker Compose, Nomad
	CI/CD	GitHub Actions, GitLab CI, Jenkins, ArgoCD
Observability	LLM monitoring	Langfuse, Phoenix, LangSmith (commercial with free tier)
	Infrastructure	Prometheus, Grafana, Loki, Jaeger
	Evaluation	DeepEval, RAGAS, TruLens, PromptFoo
Security and governance	Model security	NeMo Guardrails, Llama Guard, Prompt Armor
	Model registry	MLflow Model Registry, Hugging Face Hub (private repos)

rewrites. Maintain version-controlled model registries. Implement blue-green deployment patterns enabling zero-downtime upgrades. The AI landscape evolves rapidly; architectural flexibility provides lasting competitive advantage while specific model choices do not.

Technical metrics like perplexity and BLEU scores matter less than business outcomes. Define success in terms stakeholders care about: customer satisfaction scores, processing time reduction, error rate improvement, revenue impact, or cost savings. Connect AI system performance directly to business KPIs. This alignment

ensures continued investment and organisational support.

Open source tech stack by SLM reference architecture layer

The stack in the table above enables progressive sophistication. Organisations can begin with a simple configuration: base model, basic inference, minimal monitoring, and systematically add capabilities as requirements evolve. Each layer remains independently swappable, preventing vendor lock-in and enabling continuous optimisation.

The strategic opportunity in small language models extends beyond

current cost savings or operational improvements. Organisations building competency in SLM deployment are establishing capabilities that compound over time: proprietary training data, domain-specific model expertise, optimised inference infrastructure, and institutional knowledge of what works.

Early movers are already seeing network effects. Each production deployment generates training data for the next iteration. Each solved integration problem becomes reusable infrastructure. Each successful use case builds organisational confidence and accelerates subsequent deployments. The gap between AI leaders and